Spatial Counterpoint as a design principle in the Australia Gallery, the Melbourne Museum.

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Abstract

The Australia Gallery located on the north east of the first floor represents the shifting intersection of sound environments significant Melbourne's urban life. The diverse sound elements pertaining to the playground, the streetscape, the news stand, the Pharlap pavilion, the Markets and the MCG will be analysed and their interdependent relationships examined as a composed spatial polyphony. Particular attention will be given to the spatial design of the speaker placement as well as durational aspects of the design. The paper will be illustrated with original design information provided by John Keane, curator, and Garth Paine, sound designer and composer and Peter Jago of System Sound.

1 Introduction

Spatial Counterpoint involves the complex design of sound in many layers through an acoustic space.

The counterpoint, the line against line, layer against layer is dispersed through an entire physical space. Spatial geometries involving sound spreading, have been part of the history of sound and architecture for centuries. One is reminded of the powerful antiphonal canzonas of the Gabrielis and other spatially designed double and triple contrapuntal polychoral works of late sixteenth and early seventeenth century Italy.

Contemporary buildings and museums rarely explore the potential of their acoustic properties in clever or creative ways. All spaces are acoustic spaces whether recognized as such or not, whether acoustically designed or not. The Sydney Museum and parts of the Melbourne Museum are exceptions in that they have consciously used spatial sound as the powerful design tool it is. Both museums have considered the on effect complex layered sound designs can have for the museum visitor. This paper focuses on the way in which the Melbourne Museum has approached sound with particular reference to the Australia Gallery in the upper east galleries of the museum..

1.1 Acoustic spaces

The Australia Gallery (**Figure 1**), contains a variety of acoustic spaces in itself as a result of the architectural structure of the museum coupled with the interior partitioning from the upper floor. There is audio spill from the lower floor of the East Superspace where the sound of water lapping under the suspended Polynesian canoes can be heard from the adjoining exhibition. This gives an interesting contextual orientation to the Australia Gallery which is quite poetic. The presence of this sound encourages the posing of a question, Can one float into this space from another entry point? Indeed there is entry from the lower gallery by a staircase.

By viewing **Figure 1**, all the chief areas of the Australia Gallery can be identified. The central entrance opens into a concourse identified as the Piazza which serves as the heart of the gallery. The sound design for this space as a soundscape set the space for all the other exhibits whether isolated, (the Playground, Neighbours, the recessed Pharlap), the semi- partitioned, (the streetscapes, the archeological display) or centrally prominent, (the MCG, the sewer tour, a kiosk with news stands and a delivery van.)

From the outset, sound design was an intrinsic guiding force in the total design of the Australia Gallery in the new Melbourne Museum. Although the initial tender included multimedia, film and lighting

control."the sound led the way " according to chief curator John Keane. Garth Paine, the main sound designer of the audio content of the museum.decribes his role "I provided completed stereo mixes of the sound environments and the lighting design was done in sync to defined sections of the sounds. Those sections of the sound design in the Piazza actually come up on a semi-random order however the lighting and the temporal structure of those sections are predefined."

In my interviews with Garth Paine and John Keane it is clear that the audio was the dominant strain in integrating the various exhibits in order to give them a feeling of seamless openness, similar to being out on the street. A subtle interplay of diverse scenarios would effect this for the museum visitor.

The historian Andrew Brown -May describes the new museum as "a limitless place. With rooms never before entered.. an open book. Here there is nowhere to hide. The space of the Australia Gallery gives you the impression you're standing in the Bourke St Mall." (Brown May 2001 pp94-95). The Piazza soundscape design guidelines set criteria to ensure this was so.

2. Overall Design Guidelines

Some general principles were arrived at for the upper east piazza soundscape. These were:

1. The soundscape will be a valuable addition to the Upper East visitor experience. It provides an important way of broadening visitors 'Melbourne' sensibilities through a variety of ambient and 'key point' sounds within the Piazza space.

2. The key curatorial concern is for the soundscape not to dominate or distract visitors from their experiences within the exhibition segments themselves.

3. High points should be well spaced apart.

4. There should be room for 'quiet time' with care taken to avoid aural visitor fatigue.

5. Where possible, sounds could be related to what is in the exhibitions, especially particular objects

6. Sounds should relate directly to Melbourne street/outdoor spaces. This should provide useful parameters for focussing the soundscape and avoid trying to do too much.

7. Sounds can be recreated unless it is a voice which requires historical authenticity eg voice declaring Olympics open or Menzies declaring war.

8. These focuses were to be applied in specific and generalized ways. For example the Flinders St station would also allow for a generalised focus of transportation (i.e.. air planes - transportation - cars - trucks etc The Vic Market would allow for a generalised focus on multiculturalism in Melbourne (ethnic languages/music etc)

9. The Weather was a constant element of the soundscape. It can be perceived as subliminal most of the time, but at times it takes the foreground in rain squalls, thunder or lightning)

10. There was an attempt to historicise the sound sources of the specified sites for as many historical periods as possible i.e.1920, 1940, 1960, 1980, 2000. The soundscape progresses through these as the main thread of the soundscape. It is composed in such a way that the sound sources are perceivable individually but also as one layer in the many layers, evoking the generalised experience of the site over time.

The temporal layers of history are here being treated as spatial counterpoint.in a thoughtful way. The content of the sound design as the weaving of aggregates is the craft of the composer and sound designer. How it is dispersed spatially yields another perceptual layer of this sensitivity so that the planned complexity is elegant rather than dense to the ear. The spatial counterpoint is a principle at all stages of the design, its content and its implementation. Gradually the design process became ever more explicit in the ways this was to be achieved as it was expressed in the following document provided by the design team.

2.1 Specific Sound Design Criteria

The following Piazza Soundscape Design Guidelines were the specific application of the general guidelines and directions .

1. The focus of the sound palette is to be out-Door public spaces
2. The soundscape is not to actively deliver historical events (i.e.. speeches)
3. The Piazza sound level should approach natural outdoor public space levels occasionally
4. The Piazza should be a space in which people can sit and contemplate
5. The Piazza soundscape should not be intrusive or demanding
6. The boundaries of the locations will be bound by the CBD (i.e. not the beach or docks)
7. Sounds will be sourced from within living memory (not before 1920)

8. The soundscape is not to act as a vehicle for Sound Ecology

The soundscape will be structured into five focuses.
1. Train Stations - focusing on Flinders St and Spencer St stations
2. Markets - Vic market
3. Sporting venues - i.e. MCG
4. Parks and Gardens - Treasury gardens
5. Industry

2.2 The Sound Source Archive

Specific ideas and thousands of sound sources were discussed, recorded and collated in an archive under the general headings of sport, transport, festivals, media personalities, commerce, cultural icons, big news, past cultural events, establishment, immigration/multiculturalism and other. Other included the calls of a Herald Boy, the milkcart, a military gun. The final selection of sound sources is itemised by Garth Paine as follows:

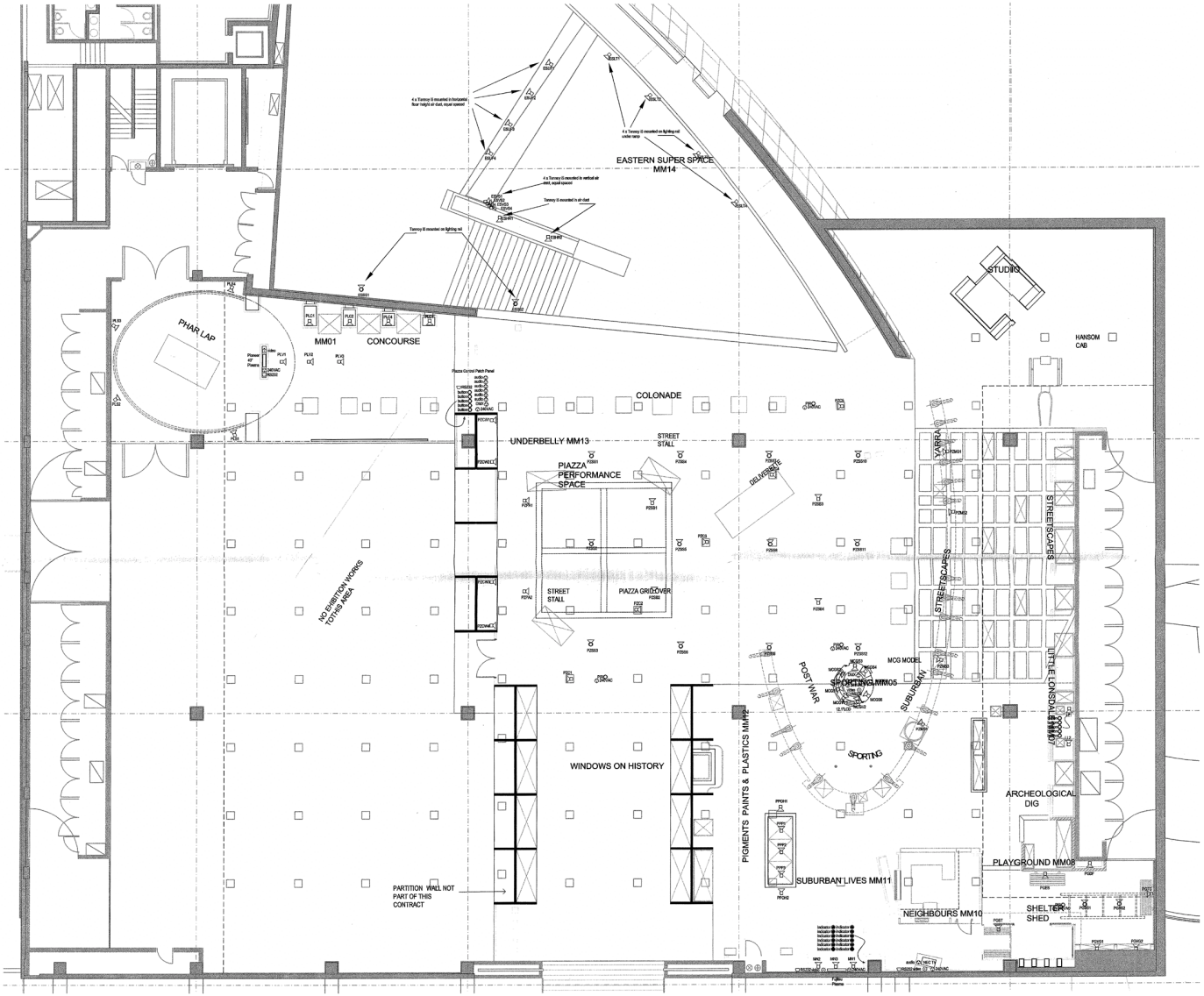


Figure 1 Floor Plan, Upper East Galleries

' Copyriight System Sound. Credits to Johns Scandrett, Garth Paine, Peter Jag and Ian Stevenson

The categories listed below were thought to best represent the experience of public space, and to present sufficient breadth of potential sound sources to make the Piazza soundscape interesting and engaging whilst presenting a quality that will allow all visitors to sit and contemplate within the exhibition space.

Location Focuses

1.Flinders Street
2.Treasury Gardens
3.Industry - Printing
4.Victoria Market
5.City Square

1. Finders Street -station and intersection

Natural Sounds
Small birds
Seagulls
Wind - clunking tins,
papers blowing,
rustling leaves
Rain - on pavement,
in gutters,
tyres through puddles,
footsteps in rain
Tyre's/footsteps in dry weather
Thunder

Man-made sounds
Trucks
Trams
Trains
Horse and cart
Bike bell - bike tyres
Boats - steam whistle
Taxis
Motorbikes
Vehicle horns
Vehicles parking, arriving, idling
Station announcements
Chit chat - various ages
varied abuse (distant yells)
Asking for and giving directions
Police whistles
St Paul's bells
Ambulance - Police - fire engine sounds
Herald boy

2. Transition to gardens

Natural sounds
Footsteps on paths
Picnics
Birds of all types
Rain on footpath
Rain on grass
Wind in trees
Dawn and dusk chorus
Dogs
Thunder and lightning

Man made sounds
Marching - parades - protest - demonstrations
Aeroplanes - flyovers Trams going past
Fountains
Children playing
Ball games (thwack thwack)
Hide and seek
Trams - cars: in groups then silence
Chit chat
Transition of accents
Shop sounds - cafes
Waiters
Sprinklers
Lawnmowers (old and new)
Rubbish collection - bin emptying
Industry (printing)

3. Industry - public space sounds

Man made sounds
Construction site
Building workers
Jack hammer - digging - saws
Timber crashing
Compressors
Cranes - whistles - dog men
Yelling - male - macho
Horses and carts
Loading of trucks
Calling out - yelling instructions -
- heads up

Fork lifts
Chunka Chunka - machinery sounds
Hisses and clunks
Tea chests - cases -
Delivery of beer
Blocks and tackles
Horse and cart harmless sound
Taxis
Air on bus and truck breaks
Reversing warning sounds

4. Victoria market

Natural sounds
Rain
Wind
Thunder

Man made Sounds
Babble
Languages
Wooden barrow wheels
Competition
Deliveries
Bartering
Arguing
Spruking

5. City square

Natural Sounds

Pigeons
Sea gull s
Footsteps on blue stone
Rain and wind

Man-made Sounds

Busking
Musical performance
Clowns - strikers
Soapbox - Speaker's corner
Passing chatter
Kids play
Skateboards
Distant traffic
Distant city sounds
Fountain
Steps - walking upanddown
At this stage it can be seen that many decisions have been made regarding the assignment of sound to location.

3 Sound Placement

The task of refining and locating the material through the space continued as a collaborative process. Personnel involved in the architectural planning, speaker design and systems control provided consultation and input. Sound placement had to integrate the acoustic limitations of the areas of the gallery, the building and the speaker placement and be compatible with the overall systems network of the museum.

Apart from the Piazza soundscape, two other spaces impacting upon the concourse had to be worked around as they involved separately designed sound as part of their exhibits. One was the partitioned and separate Pharlap room, an independent shrine/mausoleum composed as a separate audio work, yet having a video with sound at its entrance. Once inside the soft velvet curtained environment, one is acoustically partitioned from the rest of the outdoor piazza, due to the subdued absorbent material providing the regal splendour of the race track, establishing a more independent listening field. It is a place where one can focus for longer on the sonic material composed into the set as an independent theatrical composition of racing history. Here one can ponder the poetic significance of Pharlap as a national icon through the subtlety of the soundspace. The video in relation to it comes and goes providing a reference point of the actual rather than the revered world.

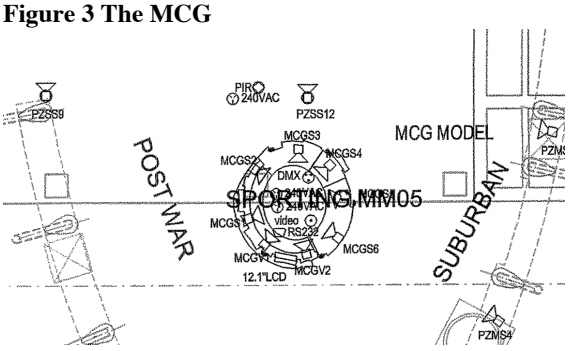
Figure 2 Pharlap



The second space, the MCG impacts more directly on the soundscape as it has a more central position as an object exhibit, the MCG arena Its bursts of distant cheering pierce the soundscape and had to be integrated to achieve the desired perspective and temporal interruptions.

The principal sounds of sport being specific to the sport, or generalised crowd noises would not lend any useful qualities to the overall ambience of the Piazza. It was felt that the sounds of sport are more related to individual activity than the experience of public space, and so became focused at the MCG arena. The design of the timing and volume of these two intrusive sound fields, The MCG and the Pharlap video had to be skillfully integrated with the overall subtler soundscape design., so their overlapping was not destructive to other material.

Figure 3 The MCG



One of the most successful areas of the Australia Gallery is the children's playground and shelter shed. Made of the materials and to the scale familiar from shelter sheds of most state schools, the area already boasts its own graffiti, proof of the public's acceptance of it. The sound, the materials, the furnishings and location in the furthest corner of the gallery, all contribute to its charm and make possible the nostalgic effect enjoyed by so many identifying the contents of their own school lunchboxes as historical cuisine. The intimacy of the acoustic and the playground stories coming and going are designed with successful randomness and unpredictability of a real schoolyard.

3.1 Speaker Placement

As can be seen from the overall floor plan **Figure 1** every attention has been given to concealed and diffuse speaker positioning to ensure the seamless spread of sound throughout the space to give the most realistic and poetic results.

The placement reveals a thorough consideration of the height, depth and width of the acoustic space and a knowledge of the changing relationships possible between the speakers.

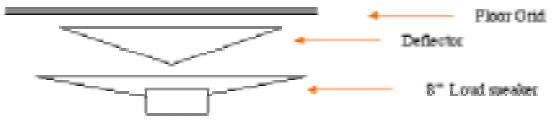
The following description provided by Garth Paine is the result of the continuing collaborative process of the larger team and it reflects the thought and detailed analysis of the physical space, a sensitivity essential to the success of this installation.

"In the Piazza for the Melbourne Museum, the spatialisation is generated using speakers in the floor, at about 100mm above the floor, at about 1600mm, and in an array through the roof.

Thirty-six loud-speakers were placed through the space. Six loudspeakers were mounted in the floor, forming a diagonal across the space. These speakers were placed about 200mm below floor level and were covered with a stainless steel cone to deflect the sound sideways, and then by a heavy grid to protect them. The floor loudspeakers were designed to carry sounds that would ground the environment. These included footsteps on pavements and gravel paths and the sound of bicycle tyres on gravel. These sounds were also spread slightly on higher loudspeakers to make the sound occupy the space even though it were 'grounded'. The floor loudspeakers were also used to pull the apparent height of some sound images down. For instance, the sound of a sprinkler running in a public park was placed in the low wall speakers and the ceiling speakers, but was primarily allocated to the floor speakers. The use of the floor speakers grounded the sound image, and the addition of small amounts of the signal in the low wall speakers lifted the sound image slightly. The addition of a very small amount of the sound image in the 1600 mm height speakers creates the sense of space experienced when hearing one of these sprinklers in an outdoor setting where the reverberation is minimal. The sprinkler sound

being sent to the 100mm height speakers was also spatialised, so that the sound moved around the space in the manner of a large public sprinkler. This created a dynamic sense of movement, the kind that makes you look to see if you are about to get wet!!

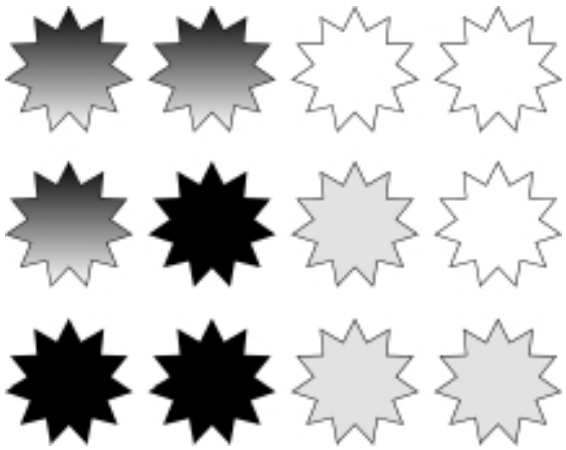
Figure 4 Speaker Mountings



A further 6 loudspeakers were placed in the kick plates of the exhibition stands approximately 100mm above floor level. These loudspeakers were mounted to face directly out into the space, and were designed to create a sense of space in the sounds emanating primarily from the floor speakers, and assist in the adjustment of the vertical position of the sound image from the ceiling and 1600mm speakers. These speakers were placed, two at either end of the space and one along each of the sides.

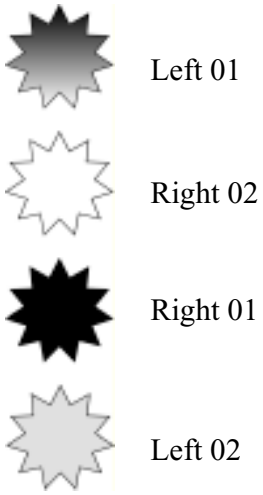
The 1600mm speakers were placed at the top of the exhibition cases. A total of four of these speakers were placed, two on each side of the exhibition space. They were used to help lift the lower sounds when I didn't want to use the overhead speakers. They assisted in the creation of *space* in the sound image.
The overhead loudspeakers formed an array of four groups of three, placed in a triangle formation as follows:

Figure 5 Speaker Groupings



This arrangement meant that it was possible to move the sound image either side from the corner. Each group took one side of the stereo image as follows:

Figure 6 Speaker Assignment



This arrangement allowed for a passing of the signal in a circular pattern, and allowed the sound to moved down the side of the array and into the centre."

From the perceiver's point of view, this information is hidden but its presence creates a spatial theatre of coming and going, or being close up and far away, so that sound is felt as proximity, dimensionality and moving matter. For example, the concealed speaker in the wall of the basket- ball ring adjoining the playground makes you look up to the real source of the sound as though the ball just missed you. One is unaware of the presence of artificial sound allowing a sense of immersion to perceive the subtle fluctuations of sound in the space.

An elegant extension of innovative speaker placement is also present on the way in to the east gallery via the escalators where one hears the windmills, water and birds as if entering a large actual , not simulated agricultural space. Speakers are carefully placed at the back of the windmill structures facing the window panes, causing an extra dimension of sound reflection. The spatial listening distance is considerably enhanced.

4 Temporal and compositional design

Garth Paine's prowess as a composer of elaborate computer music systems was fully engaged to

implement the results of this consultation in a creative and artistic way..He encoded hundreds of sound files, shaped them, morphed them and organised them into temporal aggregates which would move through the space according to the assigned areas and designed speaker placement. Although no audience interactive triggers or systems were employed, the design has inbuilt fluidity so that the various units come together at different times in different ways with varying degrees of change. All hardware systems, the spatial features of the audio tool box, the computer encoded temporal systems and the general museum networks are seamlessly integrated to effect the intention. The Australia Gallery is a fine example of sound design where many layers of subtleties unfold from different areas at different times. The result is that a space is experienced as if wandering through a genuine piazza, into an arcade, into a theatrical racing retreat, or into the school yard. The richness of the experience is sourced from the contrapuntal layers, each carefully designed as a continuum, but each having many possible aggregate lives.

It is a sound design which emulates the natural world, a sound world which comes and goes, engulfs, confronts, disrupts, floats, disappears, sustains with the ebb and flow of natural activity. The ear pulls the strands together so that each visitor has his/her own experience as they define their listening pathways and the length of stay in each location. Walking out of the Australia Gallery one is aware of having been somewhere, not quite real; but somehow familiar, engaging and immersive, integrating the past and present.

The spatial counterpoint works at many levels, the physical integrations of many parts to the whole, the layering within each particular sounding area, the intersection of each part against the other, the temporal shifts from past to present and the dramatic interplay of the elements as changeable as the weather. John Keane's commitment to sound as the focus and vehicle to set this theatricality in motion was entirely successful.. Time based systems such as these are all too rare but their power to move is readily apparent.

5 Conclusion

The Australia Gallery perhaps represents one of the most powerful and dedicated sound designs in the state of Victoria at the present time. The subtlety of the experience is achieved through the integrity of the planning, at every stage engaging spatial counterpoint as the first design principle. Layer intersects layer in four dimensions, height width, depth and in temporal shifts of small and large, modern and nostalgic, moving at different speeds, creating shifting universes. Its power creates the physical, temporal, sculptural and emotional elements of this theatrical world, entered by the player. It is a credit to the entire team that this process achieved what it has.

It is only rivalled by the Museum of Sydney, an entirely sound designed building where every gallery relates to every other on a temporal and sonic level from the outside in and the inside out, (Bandt, pp1-23.)

From understanding and experiencing clever and sensitive sound designs such as these, the public will come to demand greater input of spatial sound design in public spaces in the future.

6 Acknowledgments

Thanks to Garth Paine, composer, John Keane, Curator, Peter Jago of System Sound who provided information, interviews and discussion about their involvement in the Australia Gallery. Thanks also to Iain Mott for scanning and video editing.

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